

Assessment Task No. 4

Part I. Conceptual Understanding (20 points)

1. Define non-monotonic reasoning in your own words.

Non-monotonic reasoning is a type of logical reasoning that where a conclusion can be altered whenever a new information is presented or available.

2. How does non-monotonic reasoning differ from monotonic reasoning?

In monotonic reasoning, conclusions are independent, meaning, they are not affected whenever a new information becomes available. Non-monotonic reasoning, on the other hand, allows conclusion to alter whenever new information is presented.

3. Give a real-life situation where a conclusion must change after new information is added.

- For example, you assume it will be sunny today because the sky is clear. Later, you hear a weather update saying a storm is approaching, so you change your conclusion and decide to bring an umbrella.

4. What is a default rule? Provide one example.

*- A **default rule** is an assumption made in the absence of contradicting information. Example: "Birds can fly", this is true by default unless we learn the bird is a penguin or injured.*

5. How do argumentation frameworks help AI systems decide between conflicting rules?

*- **Argumentation frameworks** help AI systems handle conflicting rules by modeling arguments that can support or attack each other. The system evaluates which arguments are stronger or more justified, allowing it to choose the most reasonable conclusion.*

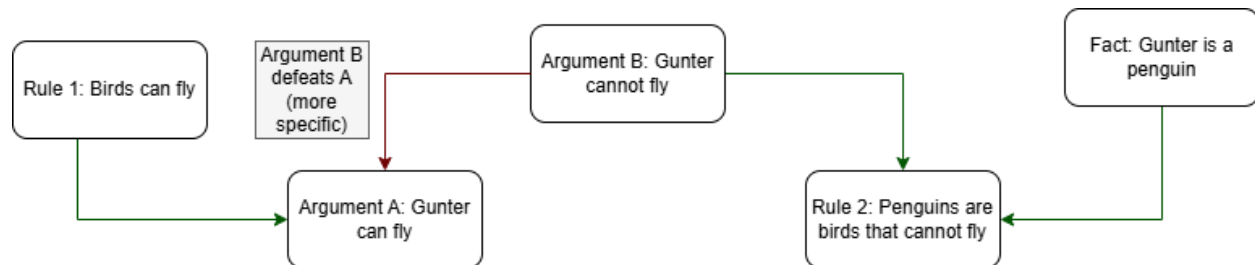
Part II. Laboratory Application (40 points)

Task 1: Belief Revision Simulation (20 points)

Program Output:

```
-----  
Input: animal = 'Penguin'  
Starting rule: If an animal is a bird, assume it can fly. (default assumption)  
User classification: This animal IS a bird.  
Default assumption applied: Assume the bird can fly (unless there is evidence otherwise).  
Assumed conclusion: 'Penguin' can fly.  
New information check: 'Penguin' matches known flightless bird(s).  
Belief revision: Withdraw the previous assumption because of the exception.  
Revised conclusion: 'Penguin' cannot fly (exception to the default rule).  
  
--- End ---
```

Task 2: Argumentation Framework (20 points)



Part III. Reflection and Discussion (20 points)

After reading Atomic Habits by James Clear, I changed how I viewed self-improvement. I used to believe that achieving big goals depended on motivation and willpower. However, after learning how small, consistent habits compound over time, it changed my perspective and conclusion: lasting change comes from systems, not sudden effort. This shift reshaped my daily mindset. I now focus on small actions that build the foundation for future growth.

This experience mirrors non-monotonic reasoning in AI, where conclusions are updated when new information becomes available. Like an AI system that revises its belief after receiving more accurate data, I replaced my old assumption with a better, evidence-based one. In both cases, reasoning is not fixed; it adapts to change. Just as AI adjusts its logic to improve decision-making, I adapted my perspective to align with new understanding. Non-monotonic reasoning, whether in humans or machines, reflects the ability to learn, evolve, and refine beliefs through experience.